

Containerization of Serverless Workloads and Understanding Its Cost

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Problem statement and Motivation

- Problem statement: Serverless computing offers scalability and cost efficiency but faces challenges like cold starts and vendor lock-in. Containerization provides portability and flexibility. This project compares the cost and performance of both approaches to identify their strengths and best-use scenarios.
- Motivation: Understanding the trade-offs between serverless computing and containerization is crucial as both are widely used in modern application development. Comparing their cost and performance helps identify the best approach for tasks requiring scalability, flexibility, and efficiency.

Background

- Serverless Computing: Services like AWS Lambda and Azure Functions offer auto-scaling but limited runtime environments.
- Containerization: Tools like Docker and Kubernetes provide isolation and consistency, differing significantly from virtual machines.

Tools/Frameworks

- Cloud Platforms: Cloudfab (a testbed for researchers where the control software is open source)
- Containerization: Docker, Kubernetes
- Serverless Platforms: AWS Lambda, Google Cloud Functions
- Cost Analysis Tools: can create a whole environment using OpenStack then monitor resource and cost usage.
- Testing of Heroku like platforms which are in the middle of both containers and serverless (a future prospect / reach goal).

Final Outcome

- A comprehensive comparison of cost and performance metrics.(WIP)
 - Execution time.
 - Resource usage (CPU, memory).
 - Cost per request or task.
- Demonstrable code for deploying and maintaining serverless and containerized workloads.
- Insights into scenarios where containerization excels.

Goal

- Minimum Goal: Compare execution time, resource usage and the resulting costs between serverless and containerized workloads.
- Reach Goal: Identify advanced use cases and check the appropriate deployment criteria (serverless or containersized). Explore multi-cloud deployments for optimized solutions.